USER STORY QUALITY EVALUATION

The INVEST framework assessment

1. Evaluation Methodology

To ensure that the elicited requirements are well-formed, feasible, and rigorous, all nineteen user stories were evaluated against the **INVEST** criteria:

- **Independent (I):** Can the story be developed and tested with minimal coupling to other backlog items? Dependencies that reflect natural workflow sequencing (e.g., booking prior to rating) are noted, whereas hard technical blocks that impede separate scheduling are flagged.
- **Negotiable (N):** Does the story capture the underlying stakeholder need and business constraint without prematurely prescribing user interface mechanics or implementation technology?
- **Valuable (V):** Is the business or user value clearly articulated and directly traceable to elicitation findings (e.g., stakeholder interviews, Sanad Care Charter)?
- **Estimable (E):** Is the story sufficiently unambiguous, bounded, and supported by explicit domain rules for developers to gauge the implementation effort?
- **Small (S):** Is the story appropriately scoped to represent a manageable work item, rather than aggregating multiple distinct triggers, user actions, or operational lifecycles?
- **Testable (T):** Are the acceptance criteria clear, verifiable, and comprehensive, covering expected behavior as well as alternative paths and exceptional/error conditions?

2. Master INVEST Evaluation Matrix

The table below summarizes the quality evaluation across all nineteen user stories. Stories flagged under **Small (S)** reflect high-level stakeholder epics; their structural decomposition and architectural rationale are detailed in Section 4.

Story ID	I	N	V	E	S	T	Overall Verdict
US-01	Pass	Pass	Pass	Pass	Pass	Pass	Pass. Cohesive data setup transaction.
US-02	Partial	Pass	Pass	Pass	Fail	Pass	Epic. Scoped at lifecycle level; expiry rule formalized.
US-03	Partial	Pass	Pass	Pass	Pass	Pass	Pass. Workload rule clarified across verified/scheduled hours.
US-04	Partial	Pass	Pass	Pass	Fail	Pass	Epic. Combines verification codes with narrative reporting.
US-05	Partial	Pass	Pass	Pass	Fail	Pass	Epic. Multi-domain dashboard; metrics and query windows bounded.
US-06	Partial	Pass	Pass	Pass	Pass	Pass	Pass. Focused financial ledger query.
US-07	Partial	Pass	Pass	Pass	Fail	Pass	Epic. Combines payroll verification with HR compliance tracking.
US-08	Partial	Pass	Pass	Pass	Pass	Pass	Pass. Interactive dialog; distress response path added.
US-09	Partial	Pass	Pass	Pass	Fail	Pass	Epic. Combines timer pause and state toggle; fall behavior defined.
US-10	Partial	Pass	Pass	Pass	Fail	Pass	Epic. Spans three discrete event triggers; daytime window bounded.
US-11	Partial	Pass	Pass	Pass	Pass	Pass	Pass. Read-only dashboard

							with timezone conversion.
US-12	Pass	Pass	Pass	Pass	Pass	Pass	Pass. Independent asynchronous media transmission.
US-13	Partial	Pass	Pass	Pass	Fail	Pass	Epic. Combines priority calling sequence with field dispatch.
US-14	Pass	Pass	Pass	Pass	Pass	Pass	Pass. Single booking transaction; temporal validations tightened.
US-15	Partial	Pass	Pass	Pass	Pass	Pass	Pass. Exemplar atomic story with complete edge-case coverage.
US-16	Pass	Pass	Pass	Pass	Fail	Pass	Epic. Plan enrollment bundles recurring invoicing; constraints set.
US-17	Partial	Pass	Pass	Pass	Pass	Pass	Pass. Discrete response action; timeout escalation fallback added.
US-18	Pass	Pass	Pass	Pass	Fail	Pass	Epic. Combines hardware infrastructure with real-time stream SLA.
US-19	Partial	Pass	Pass	Pass	Pass	Pass	Pass. Automated emergency integration; trigger wording harmonized.

3. Detailed Evaluations and Specification Refinements

Category 1: Internal Operational Roles

US-01: Care Coordinator | Daily Routine Configuration

- **Story Statement:** *As a Care Coordinator, I want to set up and update each member's daily routine, including wake-up time, meal times, and medication reminders, so that the system uses the member's normal routine when it monitors for unusual activity.*
- **INVEST Analysis:**
  - **I (Pass):** Independent entity; routine records can be established and maintained without waiting for physical sensor deployment.
  - **N (Pass):** Specifies the required behavioral parameters without dictating form layouts or input controls.
  - **V (Pass):** Forms the foundational baseline dataset required by ambient algorithms to detect anomalous inactivity.
  - **E (Pass):** Clear schema fields (times, dosages, frequencies); straightforward estimation.
  - **S (Pass):** Scoped to a single cohesive configuration profile.
  - **T (Pass):** AC 01.1–01.3 cover successful creation, update mutations, and rejection of incomplete medication schedules.
- **Verdict:** Story satisfies all criteria.

US-02: Care Coordinator — Informed Consent & Six-Month Reviews

- **Story Statement:** *As a Care Coordinator, I want to record each member's informed consent and complete the six-month consent reviews, so that monitoring only takes place with valid consent.*
- **INVEST Analysis:**
  - **I (Partial):** Natural workflow predecessor: active monitoring (-) cannot begin without valid consent.
  - **N (Pass):** Captures the legal and ethical requirement without mandating specific teleconferencing software.
  - **V (Pass):** Core requirement derived from the Sanad Care Charter and Egyptian legal frameworks for ambient surveillance.
  - **E (Pass):** Fully estimable once the expiration calculation rule was formalized.
  - **S (Fail):** Encompasses three distinct temporal operations: initial consent

capture, scheduled bi-annual review, and on-demand withdrawal.
    - **T (Pass):** Acceptance criteria verify states (, ,), but initially omitted the exact mathematical expiry rule.
- **Refinements Applied:**
    - *Specification Fix (AC 02.1 & AC 02.4):* The calculation of was formalized as:
      review_date = consent_date + 6 calendar months
      expiry_date = review_date + 30 days (Grace Period)
      This makes expiration deterministic and prevents premature service termination while reviews are scheduled.
    - *Proposed Sprint Decomposition:* (Initial Consent Recording), (Bi-annual Review & Expiry), and (Consent Withdrawal).

US-03: Caregiver Team Lead | Caregiver Visit Assignment

- **Story Statement:** *As a Caregiver Team Lead, I want to assign caregivers to requested visits based on district, skills, language, availability, and the ten-hour daily working-hours limit, so that members receive suitable care and no caregiver is overworked.*
- **INVEST Analysis:**
    - **I (Partial):** Operationally downstream from visit booking (). Can be tested using mock booking data.
    - **N (Pass):** States assignment constraints without enforcing automated algorithmic dispatch versus manual lead confirmation.
    - **V (Pass):** Optimizes caregiver-member suitability while enforcing labor compliance.
    - **E (Pass):** Criteria clearly define matching parameters (district, skills, language, hours).
    - **S (Pass):** Focuses on a single scheduling transaction.
    - **T (Pass):** AC 03.1–03.4 cover valid matches, work-hour ceiling violations, confirmation, and unassigned fallbacks.
- **Refinements Applied:**
    - *Specification Clarification (AC 03.2):* Clarified that daily hours evaluated against the 10-hour cap include both completed verified hours and currently scheduled uncompleted hours for that calendar day:
      Daily Allocated Hours = sum(Verified Completed Hours) + sum(Scheduled Pending Hours)
      If Daily Allocated Hours + Proposed Duration > 10.0 hours, assignment is blocked.

US-04: Field Caregiver | Visit Verification & Reporting

- **Story Statement:** *As a Field Caregiver, I want to confirm my arrival and departure with check-in and check-out codes and submit a short visit note, so that completed visits are properly verified and recorded.*
- **INVEST Analysis:**
  - **I (Partial):** Depends upon an assigned visit ().
  - **N (Pass):** Verification codes ensure authentication without requiring invasive GPS geo-tracking.
  - **V (Pass):** Provides the auditable verification trail required for billing and payroll settlement ().
  - **E (Pass):** Well-defined code authentication and text capture.
  - **S (Fail):** Conflates physical presence verification (arrival and departure timestamps) with service delivery documentation (clinical visit notes).
  - **T (Pass):** AC 04.1–04.4 thoroughly address code validation, rejection of invalid codes, note storage, and duration calculation.
- **Refinements Applied:**
  - *Proposed Sprint Decomposition:* (Visit Check-In and Check-Out Code Verification) and (Caregiver Post-Visit Note Submission).

Category 2: Internal Executive Roles

US-05: Founder and Director — Service Quality & Operational Analytics

- **Story Statement:** *As the Founder and Director, I want to review service information such as the number of false alerts, average alert response time, how alerts were resolved, caregiver family ratings, and members' activity trends, so that I can assess service quality and make informed decisions about Sanad's operations and growth.*
- **INVEST Analysis:**
  - **I (Partial):** Read-only reporting layer aggregating operational event records from alerts, visits, and sensors.
  - **N (Pass):** Highlights strategic performance indicators without mandating charting frameworks.
  - **V (Pass):** Essential for executive governance, SLA compliance, and operational decision-making.
  - **E (Pass):** Estimable once reporting intervals and metric formulas were bounded.
  - **S (Fail):** Bundles five independent reporting sub-domains (alert frequencies, response latency, resolution taxonomy, caregiver ratings, long-term sensor trends) into a single backlog item.
  - **T (Pass):** AC 05.1–05.6 verify reporting queries and empty-state handling.
- **Refinements Applied:**
  - *Specification Fix (AC 05.5):* Bounded the reporting window to standard

calendar intervals (monthly/quarterly) and formalized the wellbeing trend metric as a percentage variance against the member's established routine baseline:
Delta Activity = ((Sensor Counts in Window - Baseline Counts) / Baseline Counts) * 100%
    - *Proposed Sprint Decomposition:* (Incident Response & Escalation Analytics), (Caregiver Quality & Ratings), and (Member Longitudinal Wellbeing Trends).

US-06: Founder and Director — Subscription & Invoicing Oversight

- **Story Statement:** *As the Founder and Director, I want to review subscription and invoice records, so that I can oversee billing and see which payments are outstanding.*
- **INVEST Analysis:**
    - **I (Partial):** Reads billing data generated by subscription lifecycles ().
    - **N (Pass):** Specifies invoice ledger requirements without dictating ERP integrations.
    - **V (Pass):** Provides critical visibility over recurring revenue, cash collections, and arrears.
    - **E (Pass):** Standard relational queries filtered by payment status (, ,).
    - **S (Pass):** Focused single-domain ledger review.
    - **T (Pass):** AC 06.1–06.3 verify record viewing, status categorization, and outstanding balance highlighting.
- **Verdict:** Story satisfies all criteria.

US-07: Founder and Director — Staff Hours, Payroll & Compliance

- **Story Statement:** *As the Founder and Director, I want to review caregiver working hours, annual background-check renewal dates, and payroll records based on completed visits, so that I can oversee staff records and pay caregivers accurately.*
- **INVEST Analysis:**
    - **I (Partial):** Relies on verified visit durations ().
    - **N (Pass):** Defines payroll and vetting requirements without binding to a payroll disbursement gateway.
    - **V (Pass):** Ensures accurate caregiver compensation and strict adherence to safety vetting policies.
    - **E (Pass):** Well-defined attributes and arithmetic duration aggregations.
    - **S (Fail):** Combines bi-weekly operational payroll reconciliation with annual regulatory human-resource compliance (background check tracking).
    - **T (Pass):** AC 07.1–07.4 verify verified hours, background check expiry

displays, expired status flags, and payroll records.

- **Refinements Applied:**
  - *Proposed Sprint Decomposition:* (Caregiver Payroll Reconciliation based on Verified Hours) and (Caregiver Background Check Compliance & Renewal Tracking).

Category 3: External Operational Roles

US-08: Elderly Member — Smart Speaker Safety Check-in

- **Story Statement:** *As an Elderly Member, I want to respond to the smart speaker when it asks whether everything is fine, so that a false alarm is cleared quickly and my family is only called when I do not answer or need help.*
- **INVEST Analysis:**
  - **I (Partial):** Triggered by ambient anomaly detection rules (,).
  - **N (Pass):** Specifies two-way verbal interaction without restricting audio synthesis or voice recognition engines.
  - **V (Pass):** Core reassurance mechanism; provides direct elder autonomy and filters benign anomalies before disturbing family.
  - **E (Pass):** Five-minute interaction window with structured verbal response branches.
  - **S (Pass):** Single interactive dialogue transaction.
  - **T (Pass):** Initial criteria verified silence () and confirmation (), but omitted the affirmative distress branch.
- **Refinements Applied:**
  - *Story Wording Refinement:* Updated the benefit clause to explicitly capture distress: *"...and my family is only called when I do not answer or need help."*
  - *Specification Addition (AC 08.4):* Introduced explicit criteria for negative responses:
  - **AC 08.4 | Member indicates distress ("Needs Help")**
    Given: The smart speaker has prompted the member in Arabic.
    When: The member responds verbally indicating distress (*"No, I need help"* / *"لا، أحتاج مساعدة"*).
    Then: The alert records , sets , and initiates immediate family telephony escalation without waiting for the 5-minute timeout.

US-09: Elderly Member — Privacy Pause & Away Mode

- **Story Statement:** *As an Elderly Member, I want to pause monitoring for up to 24 hours or enable Away Mode, so that I keep my privacy and avoid*

unnecessary alerts when I want to be unmonitored or am travelling.

- **INVEST Analysis:**
  - **I (Partial):** Controls monitoring execution () and alert generation ().
  - **N (Pass):** The 24-hour pause ceiling is a strict policy constraint derived from the Sanad Care Charter.
  - **V (Pass):** Crucial for member autonomy, dignity, and privacy control, as emphasized in elicitation interviews.
  - **E (Pass):** Estimable once fall-detector behavior during pauses was formally documented.
  - **S (Fail):** Joins two structurally different operational mechanisms using "or":
    i. *Privacy Pause:* Ephemeral, self-reverting countdown timer (maximum 24 hours).
    ii. *Away Mode:* Continuous state latch, persisting until explicitly deactivated.
  - **T (Pass):** Criteria verify pause activation, capping, resumption, and Away Mode toggling.
- **Refinements Applied:**
  - *Specification Clarification (AC 09.1):* Formally documented that **hard-fall detection remains active and unsuppressed during a privacy pause**; only non-fall routine anomaly checks (e.g., room inactivity timers) are suspended.
  - *Specification Clarification (AC 09.4):* Defined "routine empty home alerts" as door sensor triggers, prolonged bed absence, or routine kitchen inactivity cutoff timers.
  - *Specification Addition (AC 09.6):* Added handling for pause expiration coinciding with an open alert, ensuring the system reverts to without triggering duplicate retrospective alerts.
  - *Proposed Sprint Decomposition:* (Temporary Privacy Pause with Automatic 24-Hour Expiration) and (Extended Away Mode Toggle).

US-10: Elderly Member — Multimodal Spoken Reminders

- **Story Statement:** *As an Elderly Member, I want to receive spoken reminders for my medication, scheduled caregiver visits, and wearing my fall detector, so that I take my medication on time, am ready for visits, and stay protected.*
- **INVEST Analysis:**
  - **I (Partial):** Medication reminders depend on routine data (); visit reminders depend on scheduling (/); wear reminders depend on pendant telemetry ().
  - **N (Pass):** Focuses on clear spoken audio delivery without constraining speaker hardware.
  - **V (Pass):** Directly addresses medication compliance and adherence to safety hardware wear.

    - **E (Pass):** Fully estimable once daytime intervals and mode interactions were specified.
    - **S (Fail):** Aggregates three separate business event drivers:
        i. Daily clock schedule trigger ().
        ii. 24-hour advance calendar delta trigger ().
        iii. Hardware telemetry state-change event ().
    - **T (Pass):** Initial criteria addressed only normal playback.
- **Refinements Applied:**
    - *Specification Clarification (AC 10.3):* Formally defined "daytime hours" as the interval between the member's configured wake-up time and 22:00 local time (default: 08:00 to 22:00).
    - *Specification Clarification (Away Mode Interaction):* Specified that medication and visit reminders are suppressed while .
    - *Specification Addition (AC 10.4):* Added an exceptional case handling visit cancellations prior to the 24-hour trigger, ensuring reminders are dequeued and no stale audio is played.
    - *Proposed Sprint Decomposition:* (Spoken Medication Reminders), (24-Hour Advance Visit Reminders), and (Daytime Pendant Wear Reminders).

US-11: Family Contact — Daily Wellbeing Summary

- **Story Statement:** *As a Family Contact, I want to view my family member's daily wellbeing summary in my local time zone, so that I can stay informed even when I live in another city or country.*
- **INVEST Analysis:**
    - **I (Partial):** Read-only layer dependent upon compiled daily activity logs.
    - **N (Pass):** Outlines required status indicators without constraining UI wireframes.
    - **V (Pass):** Delivers peace of mind to remote and expatriate family members.
    - **E (Pass):** Well-defined four-part status payload (, , ,).
    - **S (Pass):** Scoped to a single consolidated daily dashboard view.
    - **T (Pass):** AC 11.1–11.4 address data rendering, timezone localization, pending report states, and strict authorization checks restricting access to linked members.
- **Verdict:** Story satisfies all criteria.

US-12: Family Contact — Voice Messaging

- **Story Statement:** *As a Family Contact, I want to send voice messages to my family member's smart speaker, so that I can reassure them and stay in touch.*
- **INVEST Analysis:**
    - **I (Pass):** Self-contained media transmission feature; can be developed and

verified independently.
    - **N (Pass):** Specifies voice transmission without restricting mobile recording interfaces or compression formats.
    - **V (Pass):** High emotional value; mitigates social isolation for elderly members.
    - **E (Pass):** Standard asynchronous audio capture, storage URI generation, queuing, and delivery tracking.
    - **S (Pass):** Single atomic messaging workflow.
    - **T (Pass):** AC 12.1–12.3 verify audio queuing, delivery status state transitions (, , ,), and link verification.
- **Verdict:** Story satisfies all criteria.

US-13: Family Contact — Safety Escalation Telephony

- **Story Statement:** *As a Family Contact, I want to receive a phone call if my family member does not answer a safety check-in within five minutes, so that I can respond quickly when they may need help.*
- **INVEST Analysis:**
    - **I (Partial):** Triggered by speaker check-in timeout or distress ().
    - **N (Pass):** Specifies automated phone calling without mandating a specific telephony gateway provider.
    - **V (Pass):** Core safety mechanism ensuring family are notified immediately during potential emergencies.
    - **E (Pass):** Predictable sequential calling loop bounded by a 10-minute overall timeout.
    - **S (Fail):** Encompasses both automated sequential phone calling to family members and fallback dispatch of field personnel.
    - **T (Pass):** AC 13.1–13.4 verify call ordering, affirmative family resolution, unreachable escalation, and arrival timestamp logging.
- **Refinements Applied:**
    - *Specification Clarification (AC 13.3):* Formalized escalation taxonomy: emergency field caregiver dispatch resulting from family unreachability is classified as **Escalation Level 2B (Field Caregiver)**, distinguishing it from non-urgent volunteer response.
    - *Proposed Sprint Decomposition:* (Automated Sequential Family Telephony Escalation) and (Caregiver Urgent Fallback Dispatch).

US-14: Family Contact — Care Visit Booking

- **Story Statement:** *As a Family Contact, I want to book a care visit for my family member, so that they receive help with shopping, cleaning, or companionship.*

- **INVEST Analysis:**
  - **I (Pass):** Independent transaction; initiates the visit lifecycle.
  - **N (Pass):** Outlines core parameters (date, time, service category) without prescribing checkout mechanics.
  - **V (Pass):** Primary mechanism for family to schedule physical support for their relatives.
  - **E (Pass):** Standard form inputs and validations.
  - **S (Pass):** Single transactional booking operation.
  - **T (Pass):** Criteria address valid creation, incomplete requests, and authorization barriers.
- **Refinements Applied:**
  - *Specification Fix (AC 14.2):* Tightened input validation to explicitly reject temporal violations:
    Validation Rule: start_time > current_time AND end_time > start_time
    Bookings in the past or where duration is zero/negative are formally rejected with descriptive feedback.

US-15: Family Contact — Rate Completed Visit

- **Story Statement:** *As a Family Contact, I want to rate a completed visit, so that Sanad can see how well the caregiver served my family member.*
- **INVEST Analysis:**
  - **I (Partial):** Normal lifecycle dependency on a completed visit (). Testable via seeded visit records.
  - **N (Pass):** States the rating need on a 1–5 scale without dictating UI elements (stars, slider, numeric).
  - **V (Pass):** Feeds directly into caregiver quality tracking and supervision oversight ().
  - **E (Pass):** Single numeric attribute submission with bounded constraints.
  - **S (Pass):** Single atomic transaction.
  - **T (Pass):** Exemplar acceptance criteria: AC 15.1 covers the valid path; AC 15.2 enforces prerequisite state (); AC 15.3 validates boundary range (1 <= rating <= 5).
- **Verdict:** Story satisfies all criteria with zero modifications required.

US-16: Family Contact — Care Plan Subscription

- **Story Statement:** *As a Family Contact, I want to subscribe to a care plan for my family member, so that they are covered by Sanad's monitoring and care services.*
- **INVEST Analysis:**
  - **I (Pass):** Commercial onboarding agreement establishing service eligibility.

    - **N (Pass):** Focuses on tier selection and payment commitment without locking to a single payment processor.
    - **V (Pass):** Foundational commercial relationship enabling continuous monitoring and care delivery.
    - **E (Pass):** Standard subscription entity attributes.
    - **S (Fail):** Combines the customer's initial plan subscription agreement with the system's automated recurring monthly invoicing engine.
    - **T (Pass):** Criteria address plan selection, monthly invoice generation, and subscription constraints.
- **Refinements Applied:**
    - *Specification Fix (AC 16.3):* Formalized the business rule governing active plans:
    - **AC 16.3 | Enforce single active subscription per member**
      Given: A member is currently covered by a subscription with status .
      When: Another subscription request is submitted for that member.
      Then: The request is rejected, no duplicate record is created, and the user is informed that an active plan already covers this member.
    - *Proposed Sprint Decomposition:* (Family Plan Selection & Initial Subscription Activation) and (Automated Monthly Recurring Invoice Engine).

US-17: Neighbor Volunteer — Non-Urgent Alert Response

- **Story Statement:** *As a Neighbor Volunteer, I want to respond to non-urgent alerts in my district, so that members receive quick help from someone nearby without an emergency response*.
- **INVEST Analysis:**
    - **I (Partial):** Triggered by non-urgent alerts (,).
    - **N (Pass):** Allows flexible notification methods (SMS broadcast, push notification).
    - **V (Pass):** Harnesses community solidarity in Cairo neighborhoods; prevents unnecessary dispatch of emergency services.
    - **E (Pass):** Clear volunteer matching and claim workflow.
    - **S (Pass):** Single discrete response/claim action.
    - **T (Pass):** AC 17.1–17.3 verify geographic filtering, response logging, and urgent-alert restriction.
- **Refinements Applied:**
    - *Specification Clarification (AC 17.2):* Formally designated volunteer claim as **Escalation Level 2A (Community Volunteer Response)**.
    - *Specification Addition (AC 17.4):* Added an unhandled timeout branch:
    - **AC 17.4 | Volunteer notification timeout without response**
      Given: A non-urgent alert has been broadcast to verified volunteers in the

district.
When: No verified volunteer accepts the alert within 15 minutes.
Then: The system automatically re-escalates the alert to the primary Family Contact and notifies the nearest available Field Caregiver (Level 2B), recording .

Category 4: External Executive Roles

US-18: Technology Partner — Hub & Sensor Telemetry Monitoring

- **Story Statement:** *As the Technology Partner, I want to monitor the home hub's battery status, check whether sensor events are being delivered, and review the fall detector's reported wear status, so that I can identify device or connectivity issues that may affect the reliability of the service.*
- **INVEST Analysis:**
  - **I (Pass):** Independent technical infrastructure stream; decouples hardware telemetry from operational care workflows.
  - **N (Pass):** Specifies health indicators without restricting IoT data transport protocols (MQTT/HTTPS).
  - **V (Pass):** Critical for maintaining system availability and meeting hardware reliability SLAs.
  - **E (Pass):** Telemetry attributes (battery %, power status, connectivity) are standardized.
  - **S (Fail):** Encompasses three separate hardware domains: home hub operating status, individual sensor peripheral telemetry, and real-time pipeline ingestion latency.
  - **T (Pass):** AC 18.1–18.7 provide comprehensive pass/fail checks for all hardware conditions.
- **Refinements Applied:**
  - *Proposed Sprint Decomposition:* (Home Hub Power, Battery, and Connectivity Monitoring) and (Sensor Peripheral Diagnostics and Telemetry Latency SLA).

US-19: Emergency Services (123) — Automated Hard Fall Notification

- **Story Statement:** *As an Emergency Services (123) dispatcher, I want to receive an emergency notification within two minutes of a detected hard fall, including the member's recorded home address and the dispatch details, so that emergency responders can be sent to the correct location without delay.*
- **INVEST Analysis:**
  - **I (Partial):** Triggered by fall sensor hardware events ().
  - **N (Pass):** The 2-minute dispatch window is a life-safety SLA constraint.

    - **V (Pass):** Life-saving emergency intervention; absolute highest priority in the system.
    - **E (Pass):** Fixed payload (address, dispatch details) and deterministic trigger.
    - **S (Pass):** Focused automated dispatch integration.
    - **T (Pass):** AC 19.1–19.6 address timing, payload verification, parallel family notification, and line connection failure.
- **Refinements Applied:**
    - *Terminology Alignment:* Harmonized the story statement with the acceptance criteria. The original story phrased the trigger as a *"confirmed hard fall"*, whereas AC 19.1 initiates dispatch immediately upon a *"detected hard fall"* without awaiting human confirmation. The story text was updated to **"detected hard fall"**, while AC 19.6 records dispatch confirmation upon successful connection.

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4. Discussion of Story Granularity and Architectural Decomposition

4.1 Granularity Analysis: Epic-Level vs. Atomic Stories

A rigorous application of the INVEST framework revealed that nine out of nineteen stories (, , , , , , , and) fail the **Small (S)** criterion. In agile requirements

engineering, this pattern indicates that these requirements were formulated at the **Stakeholder Goal / Epic level** rather than as atomic sprint tasks.

Rather than fragmenting the user story catalog into dozens of micro-stories at this stage, the project team maintained these stories at the goal level in the primary catalog while documenting their logical decompositions. This approach is justified by three core software engineering principles:

1. **Requirements Elicitation Traceability:** User stories should directly reflect how stakeholders conceptualize their interactions. For example, an elderly member conceives of privacy control as a single capability (*"let me pause monitoring or turn it off when I travel"*), not as two distinct software sub-modules. Maintaining preserves a direct audit trail to Hajja Fatma's interview statements and the Sanad Care Charter.
2. **Domain Entity Cohesion:** Requirements such as Informed Consent () and Care Plan Subscriptions () represent cohesive domain state machines. Fragmenting initial recording, periodic review, and revocation across disconnected stories risks obscuring the lifecycle transitions of the underlying and entities.
3. **Decomposition via the Event Model:** In accordance with the course methodology, requirements decomposition does not rely solely on story slicing. In Part II of this deliverable, the **Event Decomposition Technique** systematically breaks down these epics into discrete, atomic business events (-). For instance, (Multimodal Reminders) is cleanly decomposed into three distinct event handlers:
   - Clock schedule trigger -> (Medication Reminder Time Reached)
   - Advance notice calculation -> (Caregiver Visit 24-Hour Notice Point)
   - Telemetry state change -> (Fall Detector Status Changed to Unworn)

4.2 Summary of Decomposed Sprint Backlog Items

For agile implementation planning, the nine epic-level stories are decomposed into the following atomic backlog items:

Original Epic	Decomposed Atomic	Core Responsibility & Boundary

	Stories	
US-02	US-02A, US-02B, US-02C	Record initial informed consent and video call verification metadata. Execute scheduled 6-month consent reviews and enforce expiration date. Process on-demand member consent withdrawal and trigger immediate monitoring cessation.
US-04	US-04A, US-04B	Verify caregiver arrival and departure using dual-OTP verification codes. Capture and persist post-visit qualitative caregiver summary notes.
US-05	US-05A, US-05B, US-05C	Query and aggregate alert response times, false alarm counts, and resolution paths. Compile caregiver performance metrics and family rating distributions. Calculate longitudinal member activity variance against routine baselines.
US-07	US-07A, US-07B	Reconcile verified caregiver hours and generate periodic payroll records. Audit caregiver background-check validity and flag upcoming/past renewal dates.
US-09	US-09A, US-09B	Manage temporary privacy pause with 24-hour automatic expiration. Toggle extended travel Away Mode for empty-home alert suppression.
US-10	US-10A, US-10B, US-10C	Trigger smart speaker medication announcements from the routine clock schedule. Issue 24-hour advance spoken notifications for upcoming scheduled visits. Emit daytime ambient reminders when the fall detector reports unworn/charging status.
US-13	US-13A, US-13B	Execute automated sequential phone dialing to linked family contacts in call order. Trigger urgent field caregiver dispatch upon 10-minute family unreachability timeout.
US-16	US-16A, US-16B	Execute family care plan selection, agreement recording, and active plan validation. Automate recurring monthly invoice generation and payment status tracking.
US-18	US-18A, US-18B	Monitor Home Hub power, battery reserves, and internet connectivity status. Audit individual sensor peripheral telemetry health and enforce 5-second delivery SLA.